The Green-Schur Bridge for OPAC-018

Clean Referee Edition v1.5.0

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Status: solution-candidate / external review pending.

This PDF is a compact reviewer overview. The editable manuscript source is paper/main.tex. The repository also includes exact-arithmetic audits, manifest verification, failure-mode documentation, and a GitHub Pages-ready website.

Truth Boundary

This package does not claim journal acceptance, official OPAC acceptance, arXiv acceptance, P vs NP, the Riemann Hypothesis, the Traveling Salesman Problem, or historical recognition. The scripts provide deterministic audit support; they do not replace independent mathematical referee review.

OPAC-018 Target

Let Phi be a finite crystallographic root system in a Euclidean vector space V, and let U be a nonzero Phi-subspace. The target is the containment:

```
pi_U(ConvHull(Phi)) subset kappa * ConvHull(Phi cap U), with kappa(Phi,U) < 2.
```

Green-Schur Candidate Route

The proposed route uses invariant-metric Cartan data, a metric symmetrizer for non-simply-laced systems, Schur complements for local projection control, and a Cellini-Marietti style transfer checklist for the global containment structure.

```
G = A D_epsilon;
S = G_UU - G_UW G_WW^{-1} G_WU
```

Reviewer Attack Points and Package Responses

Reviewer issue	Where addressed
Secret classification-by-cases	docs/CLASSIFICATION_FREE_DEPENDENCY_TABLE.md
False uniform asymptotic gap	docs/ASYMPTOTIC_MARGIN_CORRECTION.md
Long-root/short-root asymmetry	docs/NON_SIMPLY_LACED_METRIC_SYMMETRIZER.md
Oblique Phi-subspaces	docs/OBLIQUE_SUBSPACE_LATTICE_BOUND.md
Convex hull facet leakage	docs/CONVEX_HULL_CONTAINMENT_BRIDGE.md
Cellini-Marietti transfer ambiguity	docs/CELLINI_MARIETTI_TRANSFER_CHECKLIST.md

Asymptotic Safety Note

The package does not claim a rank-independent uniform gap below 2. The safe finite-rank target is kappa(Phi_n,U_n) < 2 for each finite n, while the margin may shrink with rank. This avoids a false proof route based on overbounding the full inverse matrix norm.

Metric Symmetrizer for Non-Simply-Laced Types

Using the convention A_ij = 2(alpha_i, alpha_j)/(alpha_j, alpha_j), the diagonal symmetrizer D_epsilon with entries (alpha_j, alpha_j)/2 gives G = A D_epsilon, a symmetric metric form. The Schur complement checks should use G, not the raw asymmetric Cartan matrix.

Audit Commands

```
python verify_manifest.py
python pure_python_exact_audit.py
python counterexample_stress_test.py
python verify_manifest.py
```

Optional SageMath command:

```
sage sage/opac18_green_schur_sage_test_fixed.sage
```

Audit Boundary

The pure-Python audit checks 32 finite Cartan systems for exact-arithmetic sanity conditions: Cartan integrality, symmetrizability, positive definiteness of the invariant metric form, and Schur complement sanity checks for single-node deletions. This is useful reproducibility evidence but not the proof itself.

Recommended Review Order

```
1. CLAIMS_AND_NONCLAIMS.md
2. REFEREE_ROADMAP.md
3. paper/main.tex
4. docs/CLASSIFICATION_FREE_DEPENDENCY_TABLE.md
5. docs/ASYMPTOTIC_MARGIN_CORRECTION.md
6. docs/NON_SIMPLY_LACED_METRIC_SYMMETRIZER.md
7. docs/CONVEX_HULL_CONTAINMENT_BRIDGE.md
8. Run audits
```

Final Status

The clean package is ready for GitHub upload, green CI checks, GitHub Pages publishing, and outside mathematical review. It remains a solution candidate until reviewed by qualified external referees.